

Fall 2025 ECEN 629 Homework Set 1 Solution Keys

Prob. 1. Compute the norms of the following vector $x = (-100, -99, -98, \dots, 99, 100)^T$.

- (a) ℓ_1 norm
- (b) ℓ_∞ norm
- (c) ℓ_2 norm

Note: Please consult Appendix A.1.2 if you are not familiar with the concepts.

Answer:

- (a) The ℓ_1 norm is

$$\|x\|_1 = \sum_{k=1}^{100} |-k| + 0 + \sum_{k=1}^{100} |k| = 2 \sum_{k=1}^{100} k = 2 \cdot \frac{100 \cdot 101}{2} = 10100.$$

- (b) The ℓ_∞ norm is the maximum absolute value of the entries:

$$\|x\|_\infty = \max_i |x_i| = 100.$$

- (c) The ℓ_2 norm is

$$\|x\|_2 = \sqrt{\sum_{k=-100}^{100} k^2} = \sqrt{2 \sum_{k=1}^{100} k^2} = \sqrt{2 \cdot \frac{100 \cdot 101 \cdot 201}{6}} = \sqrt{676700}.$$

Prob. 2. For a quadratic function $f(x) = x^T P x + q^T x + r$ where P is not necessarily symmetric, show that the gradient is $\nabla f(x) = (P + P^T)x + q$. Find the Hessian of the function. Provide details in your proofs.

Note: Please consult Appendix A.4 if you are not familiar with the concepts.

Answer:

- **Gradient:** Let $f(x) = x^T P x + q^T x + r$. The differential of $x^T P x$ is

$$d(x^T P x) = d(x^T) P x + x^T P dx = dx^T P x + x^T P dx.$$

Using the property $dx^T P x = x^T P^T dx$, we get

$$d(x^T P x) = x^T (P + P^T) dx.$$

The differential of $q^T x$ is $dq^T x = q^T dx$. Therefore, the total differential is

$$df(x) = d(x^T P x) + d(q^T x) = (P + P^T)x^T dx + q^T dx.$$

By definition, the gradient satisfies $df(x) = (\nabla f(x))^T dx$, so

$$\nabla f(x) = (P + P^T)x + q.$$

- **Hessian:** The Hessian is the matrix of second derivatives. Since the gradient is

$$\nabla f(x) = (P + P^T)x + q,$$

the Hessian is simply

$$\nabla^2 f(x) = P + P^T.$$

Prob. 3. A matrix $A \in \mathbf{R}^{m \times n}$ can also be viewed as a vector, by stacking all its n columns into a single length- mn column. Let $x \in \mathbf{R}^m, y \in \mathbf{R}^n$ be two column vectors. Show that

$$\langle xy^T, A \rangle = x^T Ay.$$

Answer: Set $B = xy^T$, then $b_{i,j} = x_i y_j$. It follows that $\langle xy^T, A \rangle = \langle B, A \rangle = \sum_{i=1}^m \sum_{j=1}^n a_{i,j} b_{i,j} = \sum_{i=1}^m \sum_{j=1}^n a_{i,j} x_i y_j = x^T Ay$.

Prob. 4 Let $B \in \mathbf{R}^{n \times n}$ be a given symmetric matrix of size n -by- n . Determine if the following set is convex: the set of all n -by- n symmetric A matrices, such that $A - B$ is positive semidefinite. If it is convex, provide your reasoning; if not, provide a counter-example.

Answer: Yes, it is convex. The set is $\{A | A - B \succeq 0\}$. It is the inverse image of the positive semidefinite cone under the affine mapping $f(X) = X - B$.

Prob. 5. A set is defined as

$$\{x \in \mathbf{R}^n \mid \|x - a\|_2 \leq \theta \|x - y\|_2 \text{ for all } y \in S\}$$

where $S \subseteq \mathbf{R}^n$.

- Determine whether the set is convex, which may depend on the value of θ . For the case that it is, provide your reasoning; otherwise, provide a counter-example.
- Let $a = (1, 0)^T$ and $S = \{(-1, 0)^T, (0, -1)^T\}$. For $\theta = 2$ and $\theta = 0.5$, sketch the resultant set, respectively.

Answer:

- When $0 \leq \theta < 1$, for each fixed $y \in S$, the set $\{x \in \mathbf{R}^n \mid \|x - a\|_2 \leq \theta \|x - y\|_2\}$ is in fact a ball. To rigorously show that it is indeed a ball, we need to convert it to the form $\|x - x_0\|_2^2 \leq r$, where x_0 and r are determined from a, θ, y here. The set in question is the intersection of all such balls, when y takes values in S . Thus the set will be convex. When $\theta = 1$, for each fixed $y \in S$, the set $\{x \in \mathbf{R}^n \mid \|x - a\|_2 \leq \|x - y\|_2\}$ is in fact a halfspace. The set in question is the intersection of all such halfspace, when y takes values in S . Thus the set will be convex.

When $\theta > 1$, the set is in general not convex. Let us consider the special case $\theta = 2$, $n = 1$, and S is singleton set. Then the set is $\{x \in \mathbf{R} \mid 3x^2 - (8y - 2a)x + 4y^2 - a^2 \geq 0\}$, which is clearly not convex.

When $\theta < 0$: if $\{a\} = S$, then this is the singleton set with only an element a ; if $S = \emptyset$, then the set is in fact $\mathbf{R}^n$; in other cases, it is the empty set. Thus in the case, the set is always convex.

- For each $y \in S$, the inequality $\|x - a\|_2 \leq \theta \|x - y\|_2$ defines a circle (or disc) in $\mathbf{R}^2$ with center depending on a, y , and radius scaled by θ . The set is the intersection of these discs for all $y \in S$.

- **Case $\theta = 2$:** The set is relatively large since the distance to a can be up to twice the distance to each point in S . The intersection forms a region roughly enclosing a and stretching toward S .
- **Case $\theta = 0.5$:** The set is smaller; only points very close to a satisfy $\|x - a\|_2 \leq 0.5\|x - y\|_2$ for all $y \in S$. The resultant region is concentrated near a .

The sketch is attached below.

Prob. 6. (a) Show that if S_1 and S_2 are convex set in $\mathbf{R}^n$, then the set sum

$$S \triangleq \{x_1 + x_2 | x_1 \in S_1, x_2 \in S_2\}$$

is also convex.

- (b) Let $S_1 = \{(-1, -1)^T\}$ and $S_2 = \{(2, 2)^T + u | \|u\| \leq 1\}$. What is their set sum?
- (c) Show that if S_1 and S_2 are convex set in $\mathbf{R}^n$, then the inverse sum

$$S \triangleq \bigcup_{1 \geq \lambda \geq 0} \{\lambda S_1 \cap (1 - \lambda) S_2\}$$

is also convex, where $\lambda S_j = \{\lambda x_j | x_j \in S_j\}$, $j = 1, 2$.

- (d) Let $S_1 = \{(1, 1)^T\}$ and $S_2 = \{(2, 2)^T + u | \|u\| \leq 1\}$. What is their inverse sum?

Answer:

- (a) This can be shown by definition. Let $y = x_1 + x_2 \in S$, where $x_1 \in S_1$ and $x_2 \in S_2$; similarly $y' = x'_1 + x'_2 \in S$, where $x'_1 \in S_1$ and $x'_2 \in S_2$. Then we have

$$\theta y + (1 - \theta)y' = \theta(x_1 + x_2) + (1 - \theta)(x'_1 + x'_2) = [\theta x_1 + (1 - \theta)x'_1] + [\theta x_2 + (1 - \theta)x'_2].$$

Since S_1 and S_2 are convex, $\theta x_1 + (1 - \theta)x'_1 \in S_1$ and $\theta x_2 + (1 - \theta)x'_2 \in S_2$, and thus $\theta y + (1 - \theta)y' \in S$.

- (b) See attached sketch.

- (c) This is again proved by definition. Let $y, y' \in S$. Then $y = \lambda x_1 = (1 - \lambda)x_2$ for some λ , where $x_1 \in S_1$ and $x_2 \in S_2$. Similarly, $y' = \lambda' x'_1 = (1 - \lambda')x'_2$ for some λ' , where $x'_1 \in S_1$ and $x'_2 \in S_2$. Then we have

$$\begin{aligned} \theta y + (1 - \theta)y' &= \theta \lambda x_1 + (1 - \theta)\lambda' x'_1 \\ &= [\theta \lambda + (1 - \theta)\lambda'] \left(\frac{\theta \lambda x_1}{\theta \lambda + (1 - \theta)\lambda'} + \frac{(1 - \theta)\lambda' x'_1}{\theta \lambda + (1 - \theta)\lambda'} \right) \\ &= [\theta \lambda + (1 - \theta)\lambda'] x''_1 \end{aligned}$$

where $x''_1 \in S_1$, since S_1 is convex. Similarly,

$$\begin{aligned} \theta y + (1 - \theta)y' &= \theta(1 - \lambda)x_2 + (1 - \theta)(1 - \lambda')x'_2 \\ &= [\theta(1 - \lambda) + (1 - \theta)(1 - \lambda')] \left(\frac{\theta(1 - \lambda)x_2}{\theta(1 - \lambda) + (1 - \theta)(1 - \lambda')} + \frac{(1 - \theta)(1 - \lambda')x'_2}{\theta(1 - \lambda) + (1 - \theta)(1 - \lambda')} \right) \\ &= [\theta(1 - \lambda) + (1 - \theta)(1 - \lambda')] x''_2, \end{aligned}$$

where $x''_2 \in S_2$, since S_2 is convex. It remains to check that

$$[\theta \lambda + (1 - \theta)\lambda'] + [\theta(1 - \lambda) + (1 - \theta)(1 - \lambda')] = 1,$$

which is indeed true by simple algebra.

- (d) We need to solve for λ that can satisfy this condition; see the attached sketch. This means we must have some λ to satisfy

$$(2 - \sqrt{2}/2)(1 - \lambda) \leq \lambda \leq (2 + \sqrt{2}/2)(1 - \lambda),$$

which gives $\frac{2-\sqrt{2}/2}{3-\sqrt{2}/2} \leq \lambda \leq \frac{2+\sqrt{2}/2}{3+\sqrt{2}/2}$. Thus we have

$$S = \left\{ (\lambda, \lambda) \mid \frac{2 - \sqrt{2}/2}{3 - \sqrt{2}/2} \leq \lambda \leq \frac{2 + \sqrt{2}/2}{3 + \sqrt{2}/2} \right\}.$$

Prob. 7. It is known that one halfspace contains another in the following form

$$\{x \mid a^T x \leq b\} \subseteq \{x \mid c \cdot a^T x \leq \tilde{b}\}.$$

Provide the conditions that the parameters c , b , and $\tilde{b}$ must satisfy. Are these conditions also sufficient for the containment relation between these two halfspaces?

Note: the two sets are halfspaces, therefore, must satisfy the definition.

Answer: Firstly c must be strictly positive, i.e., $c > 0$, because otherwise the two halfspaces are either disjoint, or have non-trivial set differences. Note since both sets are halfspaces, c cannot be zero here. The subset relation can be translated as follows: for any x^* such that $a^T x^* \leq b$, we must have $c \cdot a^T x^* \leq \tilde{b}$. This is true if and only if $c \cdot b \leq \tilde{b}$. Yes, the condition that $c > 0$ and $c \cdot b \leq \tilde{b}$ are both necessary and sufficient.

Note: “must satisfy” is a necessary condition, but not sufficient; example, here $c \cdot b \leq \tilde{b} - 5$ is necessary, but not sufficient. If this confuses you, it may be help to review some basic logic relation.

Prob. 8. For a pair of given sets $C, D \subseteq \mathbf{R}^n$, consider the following set

$$S = \{(a, b) \in \mathbf{R}^{n+1} \mid a^T x \leq b \text{ for all } x \in C, a^T x \geq b \text{ for all } x \in D\}.$$

Prove that the set S is convex.

Answer: For each $x \in C$, the set $S_{1,x} = \{(a, b) \mid a^T x \leq b\}$ is a halfspace. For each $x \in D$, the set $S_{2,x} = \{(a, b) \mid a^T x \geq b\}$ is also a halfspace. The set S is the intersection of these halfspaces, when taking over all $x \in C$ and all $y \in D$, i.e., $S = (\cap_{x \in C} S_{1,x}) \cap (\cap_{x \in D} S_{2,x})$.

Note that $(0, 0, \dots, 0)$ is always in S . Excluding this point, S is essentially all the separating hyperplanes of C and D , up to all positive scaling on (a, b) .

Prob. 9. Let $C = \{X \in \mathbf{R}^{n \times n} \mid a^T X a \geq 0, \forall a \in \mathbf{R}^n\}$. Determine whether the set

$$S = \{A \in \mathbf{R}^{n \times n} \mid A - M \in C\}$$

is convex, where $M \in C$ is a fixed matrix. Provide justification or a counterexample. Note that usually the definition of a matrix being positive-definite requires the matrix to be symmetric, but this problem removes that condition.

Answer:

This problem is similar to Problem 4, but we need to first establish that C is convex.

For any $a \in \mathbb{R}^n$, and any $A_1, A_2 \in C$:

$$a^T(\theta(A_1) + (1 - \theta)(A_2))a = \theta a^T(A_1)a + (1 - \theta)a^T(A_2)a \geq 0,$$

since $A_1, A_2 \in C$. Hence,

$$\theta(A_1) + (1 - \theta)(A_2) \in C.$$

Therefore, C is convex. Since S is the pre-image of the set C under an affine mapping, it is convex.

Prob. 10. Plot the norm balls of radius $r = 1$ for the following norms in $\mathbb{R}^2$ and $\mathbb{R}^3$:

- (a) ℓ_1 norm
- (b) ℓ_∞ norm
- (c) ℓ_2 norm

Answer: The sketch is attached below.